Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**HY**brid bioelectronic **P**latform for implant–wearable **E**xchange (HYPE)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Functional Restoration / Assistive & Welfare Devices	Physical Function Restoration Devices Implants, Daily Living Support Devices and Systems Mobility Support Devices
Concept of Proposed Technology	**Core Concept** The proposed technology is a **modular hybrid bioelectronic platform** combining: 1. Implantable nodes, providing bioelectronic interfaces within the body, and 2. Wearable/extracorporeal units, providing wireless power, telemetry, data processing, additional sensors and user interaction. The platform supports two architectural variants:			

- high-integration nodes with local embedded intelligence, and
- frugal nodes with minimal implanted electronics and externalized processing.

Depending on the configuration, multiple implant nodes can operate in a distributed network with synchronized or addressed communication. Wireless links (inductive/NFC/MICS/BLE) support bidirectional data exchange and wireless energy transfer, including misalignment tolerance and tissue-dependent coupling. The internal bioelectronic interface layer may be implemented either as (i) high-density thin-film electrode arrays on flexible polyimide substrates (IBMT technology), or (ii) miniaturized sensing modules supplied by the industrial partner. Beyond electrical biosignal acquisition (EMG/ENG) and stimulation, the thin-film substrate can also serve as a mechanical and dielectric carrier for further sensor modalities (e.g. electrochemical analyte sensing, impedance spectroscopy, temperature or strain microsensors).

Main Functions

Key functions of the platform include:

- Acquisition of biosignals via implantable or miniaturized sensor modules (electrophysiological, physicochemical or physical), providing a flexible interface for diverse signal types, including partner-defined sensing concepts
- Optional local actuation, such as electrical stimulation or other stimulation modalities, enabling closed-loop or interactive behavior depending on the node configuration
- Wireless energy transfer from a wearable/external unit to implanted nodes, supporting tissue-dependent coupling conditions and misalignment tolerance during operation
- Bidirectional telemetry between implant nodes and the wearable hub, allowing configuration, control, data streaming and firmware-level interaction
- Local or external signal processing, depending on whether the system uses high-integration nodes (processing on-node) or frugal nodes (processing in the wearable)
- Node networking, enabling synchronized, sequential or addressed multi-node operation and allowing more complex distributed behaviors
- Integration with wearable units, providing energy management, data processing, data storage, user interaction and potential connectivity to external devices or cloud services

Potential Uses

The platform is designed to support a wide range of non-indication-specific use cases in which implantable biosensing and external wearable processing are combined. Potential uses include:

- Biosignal acquisition for wearable analytics, where implantable or miniature sensors provide internal physiological, physicochemical or mechanical information to a wearable hub
- Closed-loop or event-triggered actuation, in which sensed biosignals can trigger stimulation or other responses depending on system configuration
- Integration into assistive or robotic systems, where biosignals (e.g. neuromuscular or mechanical) serve as control inputs for prosthetics, exoskeletons or other human–machine interfaces

	• Theranostic or research-oriented system prototypes, combining sensing and stimulation elements to enable feasibility studies without product-level commitments • Hybrid bioelectronic components for third-party devices, where electrode arrays, telemetry modules or encapsulated sensor assemblies are integrated into partner products or demonstrators These potential uses remain intentionally open and not restricted to a specific device or application, enabling SMEs from wearables, robotics, semiconductor, sensor and medical device sectors to incorporate their own sensing modalities or application concepts into the platform.
Rationale for Joint R&D Project	Hybrid implant–wearable bioelectronic systems require a combination of competencies that typically span multiple industrial and research domains. These include bioelectronic interfaces, wireless power and telemetry engineering, miniaturized sensing, packaging and biostability, embedded firmware development, wearable integration, and potentially robotics or HMI components. No single organization typically covers this entire scope, making joint R&D a meaningful and efficient approach. Fraunhofer IBMT contributes upstream R&D capabilities in: • miniaturized bioelectronic interfaces and biosignal acquisition • wireless energy transfer and telemetry link design • thin-film and MEMS-based electrode fabrication • encapsulation and biostability engineering • early validation (e.g., EMC pre-testing, bench-level biocompatibility screening) • system architecture and multi-node synchronization concepts Korean SMEs contribute downstream and product-oriented strengths, such as: • wearable and consumer-grade electronics integration • robotics, prosthetics, or HMI system development • semiconductor and sensor manufacturing • miniaturization, packaging and cost-optimized industrialization • regulatory and market pathways for regional deployment The Joint R&D project enables shared development of non-clinical demonstrators and technology transfer packages, allowing Korean SMEs to integrate their own sensor modalities, industrial design and application concepts into the hybrid platform. This collaborative approach accelerates innovation, reduces development risk, and aligns with the technological and industrial objectives of both parties.
Key Words	1. Hybrid implant–wearable bioelectronics 2. Wireless power & telemetry 3. Biosignal sensing & closed-loop actuation 4.Assistive robotics / human–machine interface 5 Wearable health monitoring / theranostics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (5 -6)
R&D Objectives	The R&D project aims to develop a modular **hybrid implant–wearable bioelectronic platform** composed of interoperable technology components. Depending on the intended application area and the competencies of the Korean SME partner, only a selected subset of the following technological objectives may be pursued within this joint R&D project. These objectives reflect typical workstreams in the development of miniaturized implantable systems. Note: Phase I of the program is used to jointly refine the target specifications and select relevant technology objectives based on SME competencies and application focus: **Architecture and system concept definition** Development and evaluation of alternative platform architectures for implantable nodes and wearable hubs, including: • high-integration “intelligent node” architectures with on-board microcontrollers and sensor preprocessing, and • minimalistic “frugal node” architectures with reduced complexity and externally managed control and processing. This objective results in a scalable technology base addressing different size, power, reliability and manufacturability constraints. **Miniaturized implantable biosignal frontend and sensor interfaces** Development of implantable frontend modules capable of interfacing with electrophysiological, physicochemical, or physical sensors. Tasks include: • thin-film or flexible substrate interfaces (e.g., polyimide, silicone) • low-noise amplification and analog frontends • optional stimulation or actuation channels (depending on configuration) **Wireless power transfer and bidirectional telemetry** Design and realization of low-power telemetry and near-field energy transfer links between implantable nodes and external/wearable units, including: • inductive / NFC-like coupling • misalignment-tolerant coil/antenna geometries • link-layer communication protocols **Multi-node networking and synchronization** Development of mechanisms enabling **distributed operation** of multiple implantable nodes, including: • addressed communication • timing and synchronization strategies • coordinated sensing or actuation events **Computational modeling, simulation and safety analysis (optional but recommended)** Computational modeling and simulation activities supporting design decisions for implantable nodes			

	and external interfaces, including: • EM field simulation for wireless power & communication • SAR & thermal modeling for tissue heating assessment • mechanical reliability and encapsulation stress evaluation **Packaging, encapsulation and biostability engineering** Development of packaging and material stacks suitable for miniaturized implantable nodes, addressing: • dielectric and moisture barrier performance • sterilization compatibility • biocompatibility screening • accelerated aging and mechanical endurance **Wearable/external control and data processing unit** Development of an external or wearable hub providing: • energy management and downlink control • data logging and basic real-time processing • test and evaluation interfaces for laboratory use **Validation, benchmarking and interoperability** Definition and implementation of bench-level validation strategies, including: • EMC pre-assessment (30 MHz – 1 GHz range) • thermal and SAR validation against simulated results • cytotoxicity screening (ISO 10993-5 or equivalent lab protocol) • interoperability testing between nodes and external hub
R&D Target Specifications	The project aims to define and achieve target specifications at platform level. The values below are indicative ranges and may be refined together with the Korean SME partner during Phase I. **Biosignal Frontend & Sensor Interfaces** • Support for multiple biosignal types (electrophysiological / physicochemical / physical) • Channel count per node depending on sensor configuration • Input-referred noise / sensitivity: suitable for tissue-level biosignals (e.g., tens of μV range for electrophysiology) • Electrode/sensor feature sizes: tens to a few hundred μm, depending on modality • Optional actuation/stimulation channels depending on node type **Wireless Power & Telemetry** • Operating distance: mm to cm range (near-field coupling) • Bidirectional data rates: kbit/s to low Mbit/s, depending on power budget • Power transfer efficiency: stable within ±20–30% coil/antenna misalignment (geometry-dependent) • Average node power consumption: compatible with continuous inductive/NFC-like powering (low-power architecture)

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Thermal, SAR & Electromagnetic Behavior

- Tissue temperature rise: target < 1–2 °C under representative operation (based on simulation + lab testing)
- EM field distribution: verified through computational modeling (FEM/FDTD) and bench measurements
- SAR constraints: evaluated via simulation tools and phantom measurements where applicable
- EMC pre-testing: basic radiated/conducted emissions assessment 30 MHz–1 GHz

Mechanical, Encapsulation & Biostability

- Encapsulation concept: decision between hermetic, non-hermetic or partly hermetic encapsulation, depending on SME use case and integration constraints
- Hermetic examples: metals or ceramics (e.g. titanium, ceramic housings)
- Non-hermetic / polymer-based examples: PDMS, silicone, parylene or comparable materials and stacks
- Target operation duration (bench): weeks to months under relevant media & temperature conditions
- Mechanical reliability: functional after representative flexural or compression cycles (application-dependent)
- Sterilization compatibility: at least one validated method (e.g. EO, low-temperature plasma, or other non-thermal techniques) depending on chosen materials
- Biocompatibility screening: bench-level ISO 10993-5 cytotoxicity for selected material combinations

Networking & Multi-node Performance

- Addressed communication: support for ≥ 3 nodes in demonstrator environment
- Synchronization: aligned sampling or triggered events across nodes
- Topology: star- or hub-based with external wearable as coordinator

Wearable / External Unit

- Energy management: stable powering of multiple nodes (e.g. ≥ 3 nodes)
- Data logging: bench evaluation from minutes to hours
- User interface: basic configuration & monitoring (GUI/HMI)
- Processing: real-time filtering & event detection for electrophysiological / physicochemical signals

Architecture Variants (Frugal vs Intelligent Nodes)

- Frugal node: minimal on-board processing, externalized intelligence → smaller footprint, higher reliability
- Intelligent node: microcontroller & preprocessing on implant → reduced bandwidth, configurable local behavior

Both variants remain within the same wireless ecosystem and can be selected based on SME-driven use cases.

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R&D Plan	The R&D Plan follows the two-phase structure of the program, consisting of a 6-month Pre-study (Phase I) and a 30-month Main R&D Phase (Phase II). The Pre-study is used to select a subset of technology objectives together with the Korean SME partner and to refine specifications based on intended use cases and manufacturing constraints. **Phase I — Pre-study (0–6 months)** Main purpose: Alignment with SME partner, consolidation of technical scope and feasibility validation of critical subsystem choices. *Key tasks / methodologies:* • T1 — SME requirement & use-case refinement: Workshops with SME to clarify signal modalities, node form factors, encapsulation preference (hermetic / non-hermetic / hybrid), and wearable constraints. • T2 — System architecture definition: Evaluation of frugal node vs. intelligent node architectures and selection based on SME capabilities and application needs. • T3 — Modeling & feasibility analysis: EM field / near-field coupling simulations, link budget analysis for telemetry/power, and preliminary thermal/SAR simulations for safety envelopes. • T4 — Technology selection & risk assessment: Material stack decision (encapsulation), sensor interface type, wireless link type, packaging approach; identification of key risks and mitigation strategies. *Expected Milestones (Phase I):* • M1 — Agreed architecture (frugal / intelligent) and encapsulation strategy (hermetic / non-hermetic / hybrid) • M2 — Refined target specifications and risk register **Phase II — Main R&D (6–36 months)** Main purpose: Development, integration and validation of selected technology components and demonstrators suitable for SME-driven productization or transfer activities. *Key Work Packages (WPs):* • WP1 — Biosignal frontend & sensor integration: Design and fabrication of thin-film/sensor interfaces; characterization of noise, impedance, sensitivity and mechanical robustness. (months 6-18) • WP2 — Wireless power & telemetry subsystem: Design of coils/antennas and link-layer protocols; lab validation under misalignment and tissue-equivalent phantom conditions. (months 6-18) • WP3 — Packaging, encapsulation & protection: Engineering of hermetic or polymer-based encapsulation; accelerated aging, mechanical cycling and sterilization compatibility testing. (months 6-18) • WP4 — Wearable / external hub development: Energy management, bidirectional communication, signal processing and data logging; basic GUI/HMI for configuration and test execution. (months 18-30) • WP5 — Multi-node networking & coordination: Implementation of addressed communication, timing synchronization and coordinated sensing/actuation scenarios across ≥ 3 nodes. (months 18-30)

	• WP6 — Validation, benchmarking & transfer: Bench-level EMC pre-tests (30 MHz–1 GHz), SAR/thermal verification vs. simulations, ISO 10993-5 cytotoxicity screening of material stacks, and preparation of transfer documentation. (months 30-36) *Expected Milestones (Phase II):* • M3 — First integrated implantable/ wearable node prototype • M4 — Demonstrated wireless power & telemetry with representative biosignal acquisition • M5 — Completed thermal/SAR, EMC and bench-level biocompatibility evaluations • M6 — Transfer package (reference designs, simulation data, test protocols) for SME-driven industrialization

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IBMT is expected to take the lead on upstream technology development and system-level research activities, providing scientific and engineering expertise across miniaturized bioelectronic systems. The expected roles and responsibilities include: • System architecture and feasibility studies, including evaluation of multiple platform variants (e.g. frugal vs. intelligent nodes) • Design and development of implantable biosignal frontends, thin-film interfaces or flexible sensor substrates • Wireless power transfer and telemetry link development, including simulations, coil/antenna design and prototype testing • Computational modeling and simulation, including EM field, SAR/thermal behavior and initial mechanical reliability assessments • Material and encapsulation research, including hermetic / non-hermetic / hybrid strategies and accelerated aging tests • Development of validation protocols and test benches for electrical, mechanical and electromagnetic performance • Bench-level biocompatibility and EMC pre-assessment, where applicable (e.g. ISO 10993-5 cytotoxicity screening, EMC 30 MHz–1 GHz) • Integration of implantable nodes with wearable/external units, including communication and synchronization schemes • Documentation and technology transfer packages to support SME-led industrialization and productization activities The research institute will provide research staff, laboratory facilities, simulation tools, prototyping capabilities and relevant testing infrastructure for the above tasks.

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Expected R&R of Korean SME Partner	The Korean SME partner is expected to take the lead on downstream development and product-oriented activities, leveraging its competencies in sensor, device, wearable or system integration. The expected roles and responsibilities include: • Definition of targeted application scenarios, signal modalities, form factors and manufacturing constraints relevant to the SME's business domain • Selection and integration of sensor modalities, including proprietary or partner-provided sensors (electrophysiological, physicochemical or physical) • Development of wearable, handheld or benchtop units, including industrial design, embedded software, UI/UX and user interaction concepts • Packaging, industrialization and supply chain considerations, including manufacturability, miniaturization and cost optimization • Device-level testing and validation, such as mechanical endurance, environmental tests, and functional evaluation under relevant conditions • Contribution to regulatory pathway preparation, if required for later commercialization (e.g. safety, EMC, reliability documentation) • Participation in system integration and interoperability testing, including multi-node scenarios and communication performance • Feedback on transfer readiness, including productization requirements, testing needs and manufacturing constraints The SME partner is expected to provide **engineering personnel, development infrastructure, local supply chain access, testing capabilities and application-specific expertise**, and may further engage subcontractors or ecosystem partners when appropriate.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to deliver validated technology building blocks for a modular hybrid implant–wearable bioelectronic platform, enabling Korean SMEs to integrate their own sensor modalities and product concepts. **Expected outcomes (technological)** • Demonstrated platform architecture including frugal and intelligent node variants • Prototype modules for biosignal frontends, wireless power & telemetry, and multi-node coordination • Simulation and verification results for EM fields, SAR/thermal behavior and key reliability aspects • Bench-level validation protocols, including EMC pre-tests and ISO 10993-5 cytotoxicity screening for selected material stacks • A transfer package (reference interfaces, test protocols, documentation) supporting SME-led industrialization Overall, the project targets a TRL increase from ~3–4 to ~5–6 through integrated demonstrators tested under relevant laboratory conditions.

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	Commercialization potential The modular platform supports multiple SME-driven pathways, including wearables and health analytics, robotics/HMI and assistive systems, and sensor and microelectronics products, with the option for later translation into regulated medical devices via separate commercialization efforts. **Broader impact** The project strengthens Korean SME competitiveness in advanced bioelectronics by reducing development risk through reference architectures and test methodologies, supporting domestic industrialization, and establishing a foundation for follow-on product development and international collaboration.

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